

Protocol — standing aphid biocontrol

Goal: no bug stuff visible on lettuce after one standard bubbler wash. Bug stuff = live insect, dead insect, parasitoid mummy, cast skin, sooty deposit.

What hinders the program's efficiency

Program-ending

- sulphur, any form, anywhere under this roof
- a broad-spectrum insecticide, pyrethrin, soap or oil

Costs most of a cohort

- refrigerating the cup
- a cup heated up (sunny dock, direct sun at the release point)
- holding past the next morning (48 h shelf life)
- releasing midday onto the top of the canopy — adults fly off toward the glazing instead of settling

Costs a share of every release

- yellow sticky cards in the cropped bays
- spraying mid-cycle instead of the day before release

Set up once

Reception

- Drop point: **indoors and shaded**. Non-negotiable!
- Never refrigerate. Room temperature, out of direct sun.
- The responsible has to be on-site on reception day.
- Someone else must be trained to replace the responsible when absent.

Order

- Cup of **50 adults** *Micromus variegatus*, weekly.

Record

- Setup pad with record sheet:
- week date
- aphid count /10 leaves
- wash-defect /10 leaves

Sprays

- Stop all sulphur, every bay, every form.

Suppliers

- NIC: \$93 for 50 adults
- Anatis: 126\$ for 50 adults on website. What about with discount?
- Supplier stock-outs happen → at the first missed week, order the following week from the other supplier (Anatis, 50 adults \$126) instead of skipping.

Every week — 10 min

1. Receive the cup. Keep it indoors, shaded, room temperature. **Never in a fridge.**
2. Quick liveness check: adults visible and moving in the cup. None moving → photo + claim a replacement the same day.
3. Release
4. At every harvest, records aphid status to close program feedback loop.

How to release

When

- **Early the next morning, first thing on arrival at work** — before the house brightens and warms.
- Never hold past the next morning (48 h shelf life).

How

1. Lay the cup open **on its side, in the shadow of the canopy, in a bed transplanted less than a week ago**
2. Leave the honeycomb paper in the cup. Do not shake, sprinkle or scatter.
3. Leave the cup in place until the bed is cut.

How to fill aphid status record

1. Before the bubbler: count aphids on **10 leaves** pulled from the beds being harvested.
2. After wash: inspect **10 washed, spun, pack-ready units**.
3. Record one line: date, aphid count /10 leaves, defect count /10 units.

Dose adjustment

Aphid count /10 leaves	Weekly order
≤ 30	50 adults
> 30	100 adults (NIC \$134.99)

- Drop back to 50 after **2 consecutive weeks ≤ 30** .

When something shows up

Trigger	Action
Colony seen during harvest or crop work	leave it — adults find colonies themselves; recount at next harvest
Count still > 30 after 2 weeks at 100 adults	order 500 <i>Chrysoperla</i> larvae → release on the hotspots, 10 /m²; inspect the nursery cohort due to transplant next
Shipment dead or missed, no replacement in time	order 500 <i>Chrysoperla</i> larvae as the bridge week
Sulphur gets applied anyway	pause the auto-ship for 3 weeks
A mildew spray is due	spray the day before release day; never mid-week